

Little League Hamilton

Capstone Project Proposal

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<https://github.com/TimothyMwangi101/Little-League-Hamilton>

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Project Description and Scope

The project will build a website for people to form teams to play sports, namely basketball, baseball, football, soccer, bocce and tennis. The website aim is to allow beginners to meet up with other beginners to play casual sports. The website will display a map with markers of the locations of city owned parks in Downtown Hamilton and Hamilton Mountain. When clicked, the marker will display what sport is played, use google street view to see the park, if there are any matches being played and if so which team and give the user directions to the map.

The software will allow users to track goals/touchdowns, wins and losses, assists and schedule matches with other teams. The website will have a dashboard that will show recent matches, upcoming matches and leaderboards. The website will optionally allow players to form and join teams, which will have a dedicated roster of players and will have additional roles of Captain for teams bigger than 3 people and Co-captains for teams larger than 7. Co-captains are responsible for scheduling matches, updating team profiles, and naming the MVP after matches. Captain has all co-captain roles in addition to adding/removing players from the team and promotion to Co-captain

Contains public sector Data made available under the City of Hamilton's Open Data Licence:
<https://open.hamilton.ca/datasets/SpatialSolutions::park-sports-fields/about>

Common Terminology

Sports – Represents the following sports only, basketball, baseball, football, soccer, bocce and tennis.

Player – Most basic user role. Represents a person playing sports.

Co-captain – A player in a team of at least seven who is promoted by the Captain.

Captain – A player with the most responsibility in a team.

Team – a group of four or more players with Captain and optionally Co-captain roles.

Parks – the place where sports will be played by players and teams.

Most Valuable Player (MVP) – the player who played the best. Can't be a Captain or Co-captain.

Score – a general name for touchdown/goals or whatever other sport specific term that represents the same concept.

Map – Google Maps

Must – word indicates the requirement is mandatory to be implemented

May – word indicates the requirement is optional to be implemented

Requirements

Functional Requirements

1. The website must allow a user to register themselves as a player.
 - 1.1. The user can access all parts of the website except the team functionalities which a user must be registered
 - 1.2. The registration must ask for the user's email address and password
 - 1.2.1. Passwords must be salted and hashed before they are stored in a database
 - 1.2.2. The password must be more than 8 characters but less than 30 characters
 - 1.2.3. If the email is already registered, then the user is asked to login instead
 - 1.3. The website must allow the user to create a new password if they have forgotten their password
 - 1.3.1. This must be a link in the login screen with a link "Forgot password?" leading to the password reset functionality
 - 1.3.2. The website must ask a user to input an email address to send a password reset message to. If the email address is not of a registered user, a message must be shown to the user, "No valid user registered with your email address".
 - 1.4. The website must have a logout function for logged in users.
 - 1.4.1. Logged out users must not have access to the team functionalities
 - 1.4.2. All information stored as cache must be deleted
2. The website must have an interactive map provided by Google Maps centered around Hamilton Downtown and Mountain area
 - 2.1. The map must display colored markers where parks are in which sports can be played
 - 2.1.1. The colors must be red, orange, yellow, green, blue, and purple for basketball, baseball, football, soccer, bocce and tennis respectively.
 - 2.2. Clicking on a marker must show an information panel.
 - 2.2.1. This panel shows what sport can be played
 - 2.2.2. This panel must have a view in google street view functionality
 - 2.2.3. This panel must show if there is any scheduled sport on that park
 - 2.2.4. This panel must have a button that shows the directions to the park when clicked
3. The system must allow a registered user to form a team.
 - 3.1. Whenever a team is formed, the player who created the team must be the Captain.

- 3.1.1. Team's must be open to join without a request or have restricted membership based on being invited by the Captain.
 - 3.2. Team members must have the ability to leave a team at will.
 - 3.2.1. If a Captain wants to leave a team, they must first promote the co-captain to captain or disband the team
 - 3.2.2. A team that has been disbanded has all their members removed and they can no longer play as a team
 - 3.3. If a team has more than 7 members, the website must have functionality that allows the Captain to promote a player to Co-captain.
 - 3.3.1. Co captains must be responsible for scheduling matches, updating team profiles, and naming the MVP after matches
 - 3.3.2. Captains must have all co-captain responsibilities in addition to adding/removing players from the team, ability to disband a team, and promotion to Co-captain
 - 3.3.3. To schedule a match, the co-captain or captain must fill a form with the following information:
 - Sport
 - Park(s) – The user must have the option to select up to 3 parks
 - Number of participants
 - Time and date of the match
 - Opponent – Then user may select a specific team that they want to match against
 - 3.3.4. Once the form is submitted, the match is published and must be visible to the selected opponent only or all other teams, if opponent not specified.
 - 3.3.5. These teams or the opponent must have an option to accept or decline the match. If all teams or the opponent declines the match, then the match is cancelled. If a team accepts, then all other teams cannot accept or decline the match.
4. The website must have a homepage that will show recent matches, upcoming matches and a leaderboard table
 - 4.1. The leaderboard must make use of pagination, by showing a maximum of 10 entries at a time and allow the user to request the next 10 entries and navigate backwards to the previous entries
 - 4.2. The leaderboard must have an ascending or descending sort feature
 - 4.2.1. The user must be able to sort the win/loss statistics, name (alphabetically), total goals and total matches
 - 4.3. The leaderboard must have a search feature. This search will enable the user to search by team name and return the searched team only

Non-Functional Requirements

1. The website's must codebase must be modular and built with the expectation that a new sport or expanding the map to allow more regions within Hamilton
2. The interactive map must load within 4 seconds
3. Passwords must be salted and hashed before storing in the DB
4. Errors must be handled gracefully. Errors will lead to its own dedicated page with an error message and status code.
5. The website must use parametrized or prepared SQL statements to prevent SQL injections
6. All sessions and cookies must use the HttpOnly, Secure, and SameSite flags for robust security
7. Images and HTML elements will make use of the alt attribute to assist users with accessibility needs as outlined in [1.4.5: Images of Text](#)
8. Buttons elements must use the same colors and all clickable elements must have a hover pseudo element as outlined in [1.4.13: Content on Hover or Focus](#)

Initial Testing Plan

Functional Frontend Testing – Using Playwright

Test Case ID	Reason	Data Input	Expected Output	Requirement Mapping
F1	Successful Registration	Email: user@gmail.com; Password: p@ssword	Account created; Player role; team features gated	F1, F1.2–F1.5 NF 3
F2	Duplicate Email Registration	Email already registered	Registration blocked; prompted to log in	F1.6
F3	Password Length Validation	7-char and 31-char passwords	Validation error shown	F1.5
F4	Password Reset (Known Email)	Forgot password flow with registered email	Reset email sent	F1.7–F1.10
F5	Password Reset (Unknown Email)	Unknown email in reset form	Error: No valid user	F1.9
F6	Logout & Cache Clear	Click logout	Session cleared; no access team functions	F1.11–F1.13
F7	Map Load Time	Measure load	<=4s	NF2
F8	Marker Colors	View markers	Colors correct and match with there corresponding sport	F2.1
F9	Marker Info Panel	Click marker	Panel shows details	F2.2

Unit Testing – Using Vitest

Test Case ID	Reason	Data Input	Expected Output	Requirement Mapping
Unit Testcase 1	Create Team	Team creation details	Creator becomes Captain	F3.1
Unit Testcase 2	Join Policy	Join open/invite team	The user joins the team if it has an open policy or are invited	F3.2
Unit Testcase 3	Leave Team	Kick member or member leaves team	Member removed	F3.2
Unit Testcase 4	Captain Leave Rule	Captain leaves	Blocked unless promote/disband	F3.2.1
Unit Testcase 5	Co captain Permissions	Schedule matches, updating team profiles, and naming the MVP after matches	Correct permissions	F3.3.1
Unit Testcase 6	Captain roles	Adding/removing players from the team, disbanding a team, and promotion to Co-captain	Correct permission	F3.3.2
Unit Testcase 7	Schedule Match	Fill the form	Match published and visible to other teams	F3.3.3
Unit Testcase 8	Visibility	Opponent specified/none	Visibility correct	F3.3.4
Unit Testcase 9	Accept match	Specified opponent accepts match	First team to accept the match gets the match. No longer visible to other teams	F3.3.5
Unit Testcase 10	Decline match	All teams/selected opponent rejects match	Match is canceled	F3.3.5

API Testing – Using Postman

Test Case ID	Reason	Data Input	Expected Output	Requirement Mapping
API Testcase 1	Homepage Sections	Access the website	Scheduling and team feature not visible	F1.1
API Testcase 2	Pagination	Navigate leaderboard table	10 entries per page	F4.1
API Testcase 3	Sorting	Sort by win/loss, name, total goals and team name	Sorted by ascending/descending	F4.2
API Testcase 4	Search	Search team by name	Only that team is returned	F4.3
API Testcase 5	Map Loads	Open Map page	Map loads centered around Hamilton Downtown and Mountain region	F2 NF2

Security Testing – Using Zap

Test Case ID	Reason	Data Input	Expected Output	Requirement Mapping
S1	Password Storage	Inspect DB	Salted+hashed	NF3
S2	SQL Injection	Try payloads	Protected	NF5
S3	Cookie Flags	Inspect cookies	HttpOnly, Secure, and SameSite flags set	NF6

Accessibility Testing – Using AccessibilityChecker.org

A1	Test accessibility for screen readers	Images and HTML tags	Alt present	NF7
A2	Hover/Focus	Hover/tab UI	Indicators visible	NF8

Timeline

NOTE: Item in blue is an example – you can remove
 Items in red is required and cannot remove

Week / Dates	Development, Testing and Deliverables To Be Completed	Requirement Mapping
Week 1 Sept 9 th – 11 th	Submit Capstone Proposal Document in MyCanvas Course Shell Attend Capstone Launch Lecture **No Development in this week	
Week 2 Sept 14 th – 18 th	Authentication Functionality Implemented Testing: Security, Front end testing for logging in	F 1.2, 1.3, 1.4 NF 3, 5, 6
Week 3 Sept 21 st – 25 th	Submit Milestone 1 (Sept 25th)	
Week 4 Sept 28 th – Oct 2 nd	Integrate Google Maps API Testing: Frontend – the panel and marker, API testing weather the maps is set up properly	F 2.1, 2.1 NF 2
Week 5 Oct 5 th – 9 th	Implement team creation, joining and leaving API testing & Frontend Testing for team creation, joining and leaving	F 3.1, 3.2, 3.3.1, 3.3.2 NF 1
Break Week Oct 12 th – 16 th	**No Development in this week	

Week / Date	Deliverables	Requirement Mapping
Week 6 Oct 19 th – 23 rd	Initial Deployment of Website Submit Milestone 2 (Oct 23rd)	
Week 7 Oct 26 th – 30 th	Implement co captain promotion and captain roles Frontend and API testing on co captain & captain roles	F 3.3.1, 3.3.2
Week 8 Nov 2 nd – 6 th	Implement Scheduling Frontend and API testing on scheduling	F 3.3.3 NF 5
Week 9 Nov 9 th – 13 th	Submit Milestone 3 (Nov 13th)	
Week 10 Nov 16 th – 20 th	Implement accept/decline matches Frontend and API testing on accepting and declining matches.	F 3.3.4, 3.3.5
Week 11 Nov 23 rd – 27 th	Implement searching, sorting and pagination for the leaderboard Frontend and API testing on leaderboard features	F 4
Week 12 Nov 30 th – Dec 4 th	Implementing error page, accessibility, unit, API, Frontend, Security testing all functionalities	NF 1, 4, 7, 8
Week 13 Dec 7 th – 11 th	Finalize Testing Submit Milestone 4 **No Development in this week	

Preliminary Deployment Research

I plan to use [Railway](#) for hosting and deployment. Railway offers instant deployment from multiple options such as from a GitHub repo and a Docker Image providing lots of flexibility and will allow me to move fast.

Railway has a free tier which offers a free domain, modest CPU, memory and storage. The build image is 4GB which may be too little. Most importantly, it allows me to set hard limits, allowing me to limit costs.

If the free tier isn't enough or too restrictive, I can opt to upgrade to the Hobby tier for \$5 a month. Additional to the free tier perks, I can get 2 custom domains, 100GB image build size, and significantly more compute resources

Entity Relationship Diagram

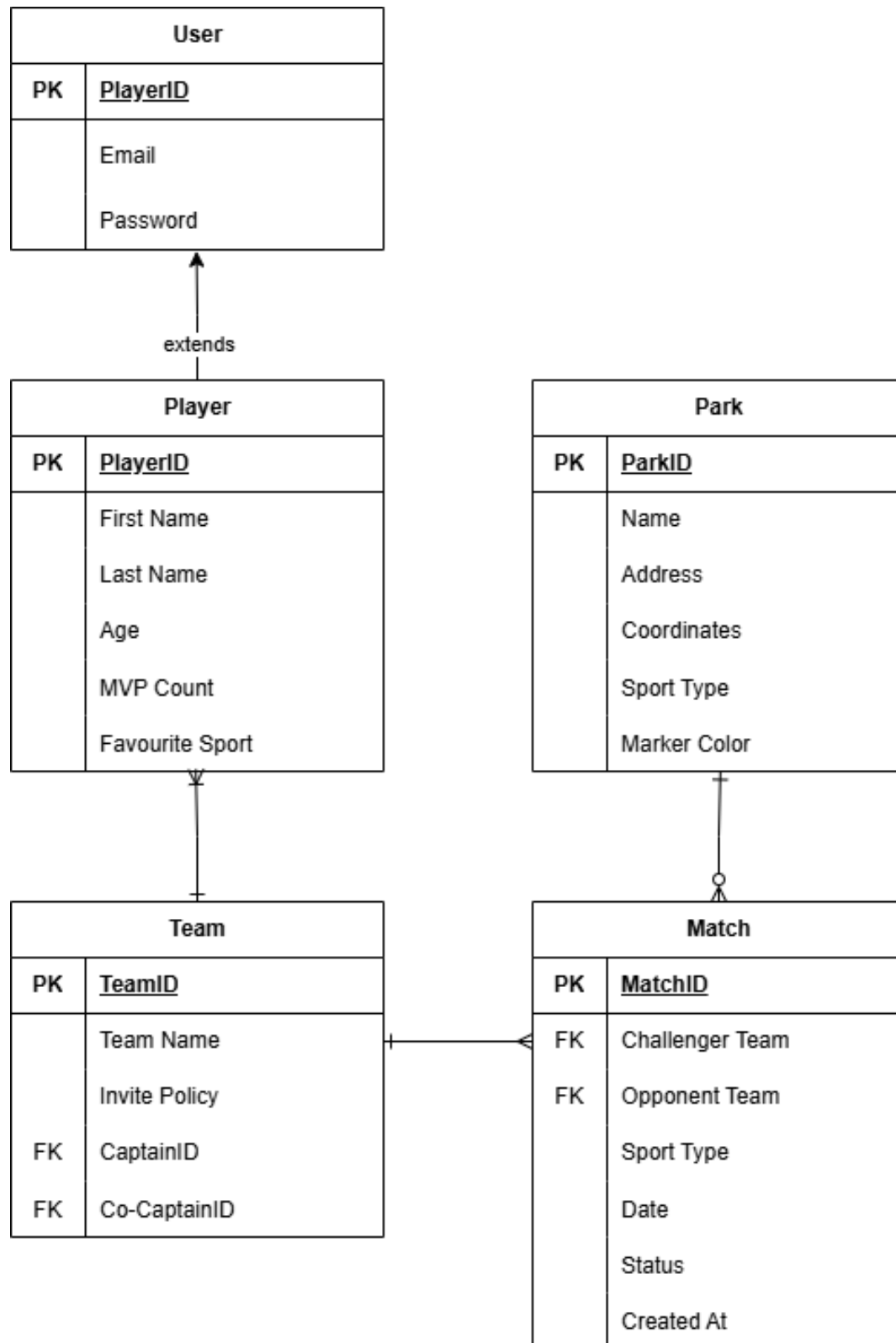


Figure 1: The ERD of the System

User Interface Mock Ups

Login Page

The login page features a dark green background on the left and a light beige background on the right. On the green background, the text 'Hamilton Little League' is displayed in a large, white, serif font, with 'Accessible and Casual sports for Hamiltonians' in a smaller, white, sans-serif font below it. On the beige background, there are two input fields for 'Email Address' and 'Password', each with a placeholder text 'Enter your email address' and 'Enter your password' respectively. Below the password field is a link 'Forgot password?'. A dark green 'LOG IN' button is positioned below the password field. At the bottom, there is a link 'Sign up here' preceded by the text 'Don't have an account?'.

Hamilton Little League
Accessible and Casual sports for Hamiltonians

Email Address
Enter your email address

Password
Enter your password

[Forgot password?](#)

LOG IN

Don't have an account? [Sign up here](#)

Figure 2: Desktop View of the Landing Page

Hamilton Little League

Accessible and Casual sports for
Hamiltonians

Email Address

Password

[Forgot password?](#)

LOG IN

Figure 3: Mobile View of Login Page

Dashboard

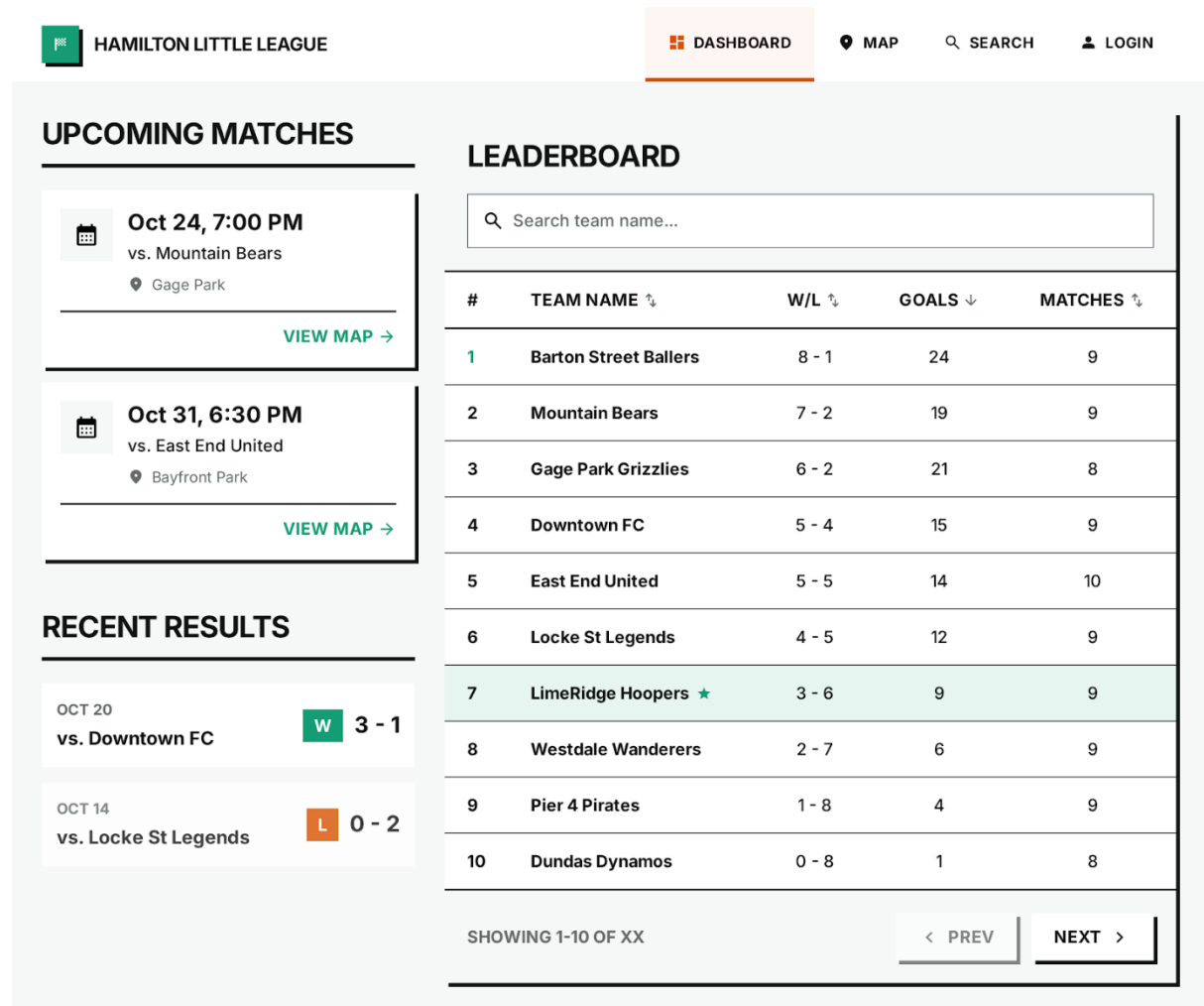


Figure 4: Dashboard view which includes an onclick sort feature on table headers

Create Team Form

← Back to Dashboard

Create a New Team

Team Details

Please fix the following:

- Select a sport
- Provide a team name

Which sport will your team play?


Basketball


Baseball


Football


Soccer


Bocce


Tennis

Please select a sport from the options above.

Basic Information

Team Name

e.g. Mountain Top Ballers

Team name is required.

Team Logo



Upload a team logo

Choose File No file chosen

Recommended: PNG or JPG, square aspect ratio, max 2MB.

Create Team →

Figure 5: Create Team Desktop View

[← Back to Dashboard](#)

Create a New Team

Team Details

⚠ Please fix the following:

- Select a sport
- Provide a team name

Which sport will your team play?

 Basketball	 Baseball
 Football	 Soccer
 Bocce	 Tennis

Please select a sport from the options above.

Basic Information

Team Name

e.g. Mountain Top Ballers

Team name is required.

Team Logo



Upload a team logo

[Choose File](#) No file chosen

Recommended: PNG or JPG, square aspect ratio, max 2MB.

Create Team →

Figure 6: Create Team Mobile View

Map View

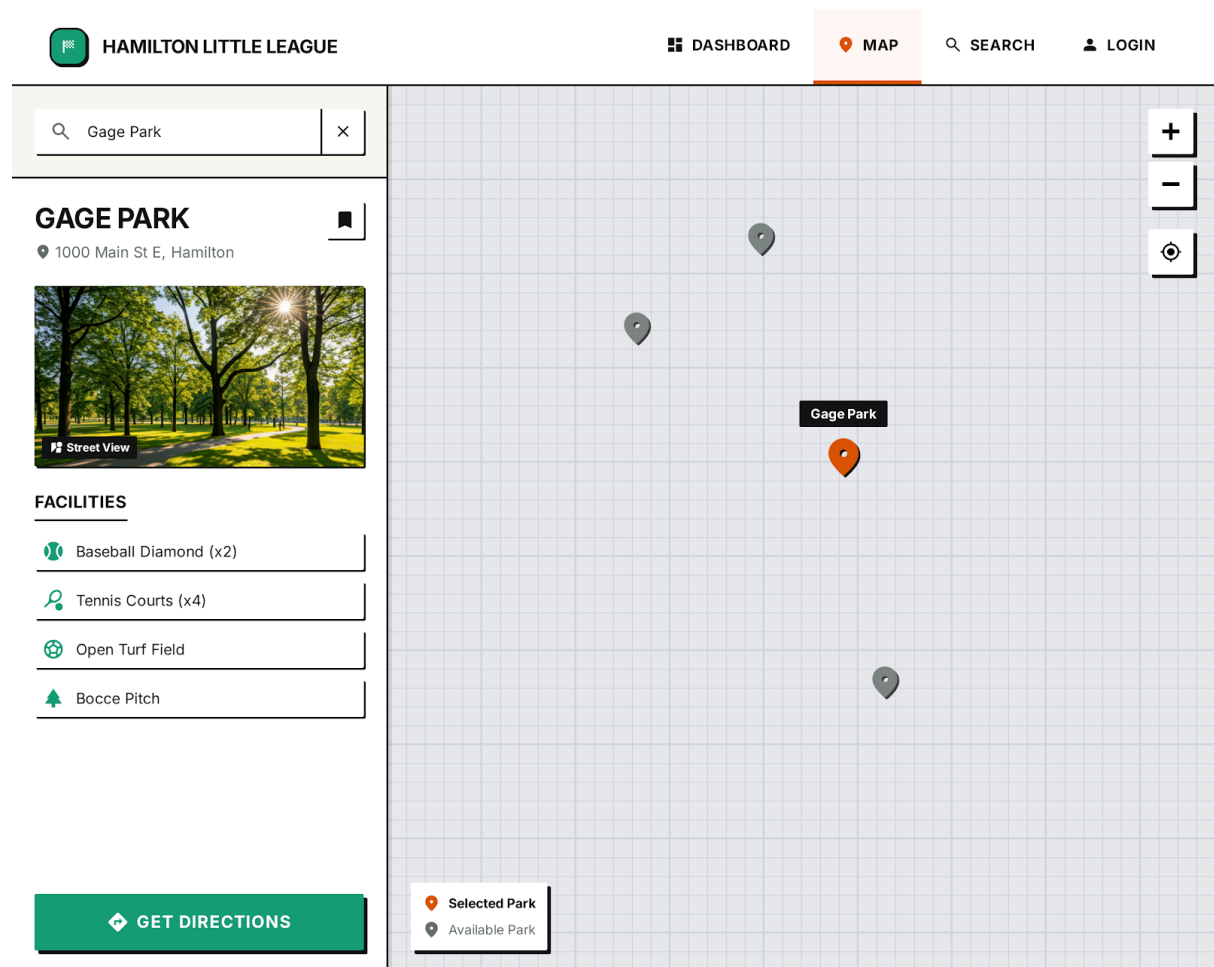


Figure 7: Map view

Error Page



Out of Bounds.

The page you are looking for doesn't exist or
has been moved off the court.

[← RETURN TO DASHBOARD](#)

Figure 8: Default Error Page

Search and Filter Page

 HAMILTON LITTLE LEAGUE

DASHBOARDMAPSEARCHLOGIN

Filters

[Clear All](#)

Sport

☒ Basketball

☐ Baseball

☐ Soccer

☐ Bocce

☐ Tennis

Region

☒ Hamilton Mountain

☐ Downtown

Basketball ×Hamilton Mountain ×

Results (12)

BASKETBALLOpen

Mountain Hoopsters

Captain: Sarah J.

VIEW TEAM

BASKETBALLOpen

The Backboard Breakers

Captain: Mike T.

VIEW TEAM

BASKETBALLOpen

Escarpment Elite

Captain: Alex W.

VIEW TEAM

BASKETBALLInvite Only

Concession Cruisers

Captain: David L.

VIEW TEAM

PREVIOUS

Page 1 of 3

NEXT

Figure 9: Filter and Search page

User Profile

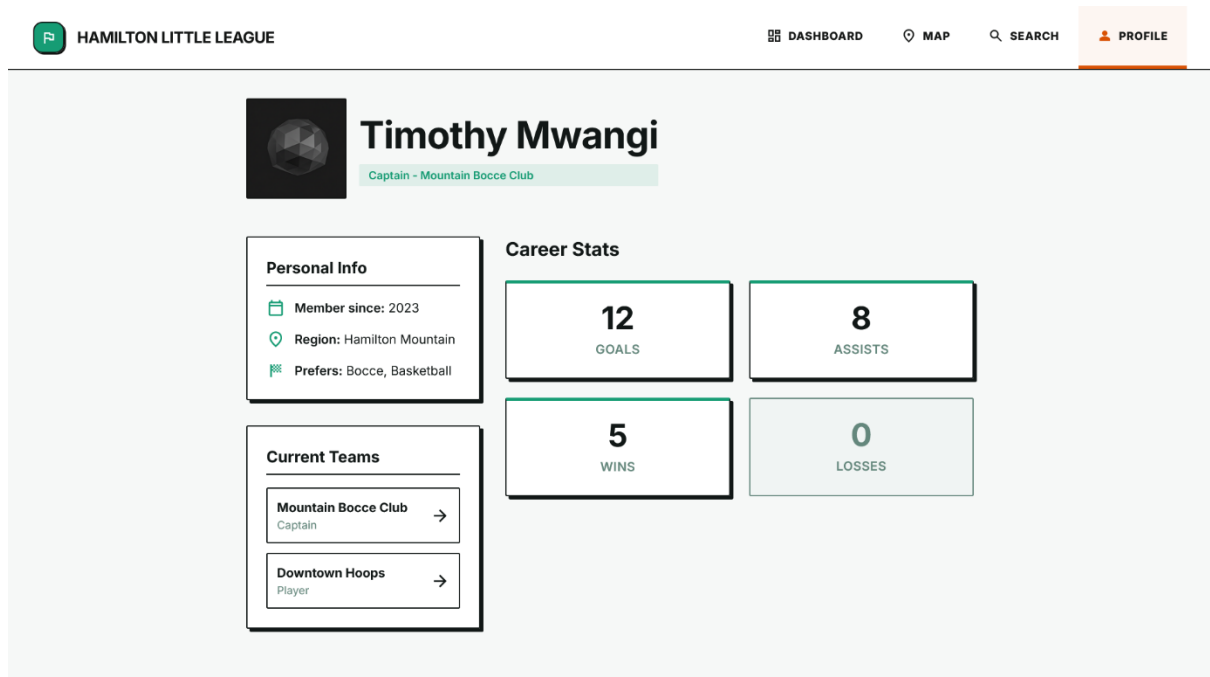


Figure 10: User Profile

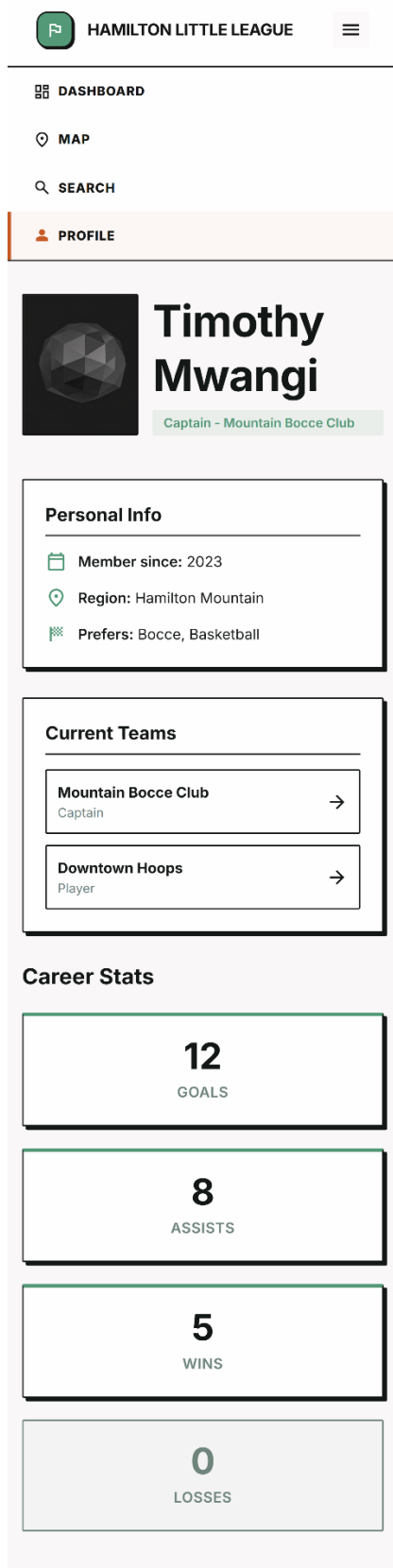


Figure 11: Mobile View showing the Collapsible Navbar